ProteinMPNN Progress Update

4/8/26

ProteinMPNN

- Determines the sequence of a protein given the backbone
- **Autoregressive decoding algorithm** places amino acids one at a time, conditioning on previously selected amino acids

$$p(x_{1:T}) = \prod_{t=1}^{T} p(x_t \mid x_{<t}) \, .$$

- **Mental model**: each residue is a node → find 30 closest neighbors → calculate edge features between each node + its neighbors using the backbone → place an AA at pos 0 (condition on backbone) → place AA at pos 1 (condition on backbone + pos 0) → ...

ProteinMPNN Metric

Primary metric: sequence recovery

- Percentage of the sequence generated by ProteinMPNN that is the same as the target/original sequence

Other possible metrics:

- Energetics
- Stability
- Similarity of the structure predicted by Alphafold to the original structure (regardless of the original sequence)

The Problem: Limitation of Autoregressive Decoding

- Amino acids earlier in the protein are conditioned are none or very few other residues
- The model never verifies these residues after completing the rest of the sequence
- **Ex**: if ProteinMPNN selects LEU at position 0 conditioning only on the backbone but, given the rest of the sequence, ALA is actually a better choice, the model never checks its original selection and changes position 0 to ALA

Early Experimentation

- Added a second iteration through the sequence to verify
  - First 10-15 AAs
  - First 10-15% AAs
  - All amino acids (now every residue is conditioned on the entire rest of the sequence)
  - Sequence recovery improvement was minimal
- Annealing
  - Create a “rough sketch” of the sequence during the first pass of ProteinMPNN using a high temperature
  - Revise the sequence through subsequent passes that condition on the entire sequence, gradually reducing temperature
  - Sequence recovery did not improve significantly

Challenges

1. Circular Dependency

- AA0 determines AA1 which both determine AA2, ...
- Since earlier residues determine later residues, those later residues will reinforce the earlier selections when we perform a second pass to verify the early AAs
- This could explain a lack of improvement in sequence recovery with an added second pass

2. ProteinMPNN has Discrete Outputs

- ProteinMPNN outputs logits that represents the raw scores for each amino acid at a certain position
- Argmax selects the most probable amino acid at that position
- ProteinMPNN outputs the argmax value
    - When we feed the outputs of one ProteinMPNN iteration back into the model for another iteration, we are using the discrete argmax values
    - There is significant information loss in converting a continuous distribution to one discrete value
    - Subsequent iterations of the model do not know the other likely AAs from previous iterations

3. Protein Breathing

- **Goal**: quantify the degree to which recovery rates are robust against physical perturbations + produce sequences that account for protein breathing
- Add a small gaussian to the model → will significantly drop recovery + rate of folding → model will output a nonphysical sequence
- Relax the sequence via **AMBER**
    - AMBER (Assisted Model Building with Energy Refinement)
- Reduce potential energy of the sequence